



# INDUCTIVE BIAS IN MACHINE LEARNING

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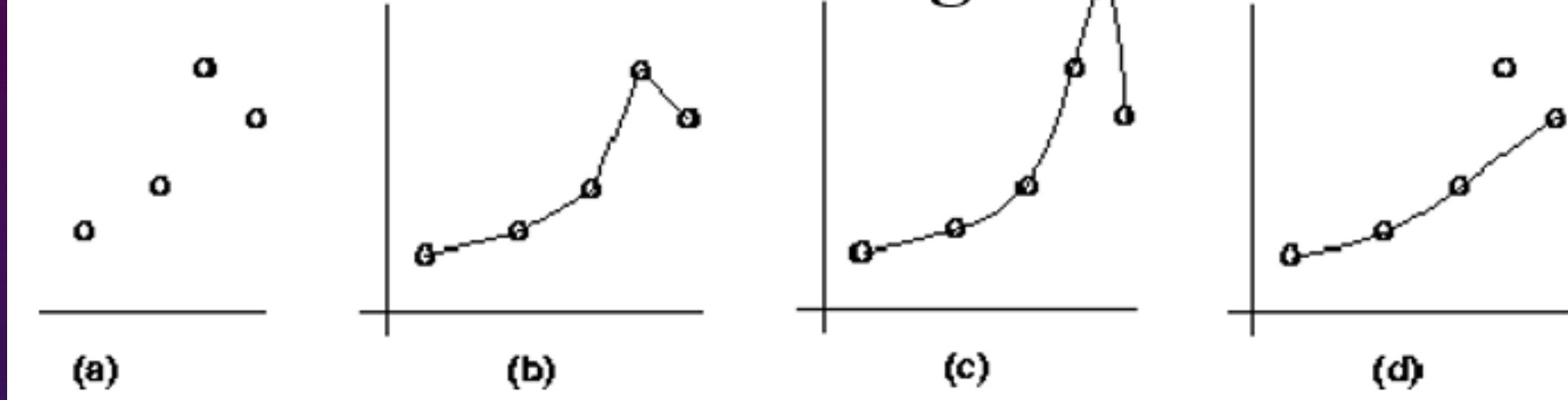
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# Inductive Bias

- Inductive bias of a learning algorithm is the set of assumption that , together with the training data ,the algorithm uses to predict outputs for unseen inputs
- Also known as learning bias
- It helps the algorithm focus on what's important in the data and avoid getting lost in irrelevant details. It steers the learning process towards specific types of solutions

# Inductive learning and bias



- Suppose that we want to learn a function  $f(x) = y$  and we are given some sample  $(x, y)$  pairs, as in figure (a)
- There are several hypotheses we could make about this function, e.g.: (b), (c) and (d)
- A preference for one over the others reveals the **bias** of our learning technique

e.g. :

- prefer piece-wise functions (b)
- prefer a smooth function (c)
- prefer a simple function and treat outliers as noise (d)

- Inductive bias is important because it affects the generalization performance of the machine learning algorithm
- A machine learning algorithm with a good inductive bias will generalize well to new, unseen data
- while an algorithm with a bad inductive bias may overfit to the training data and perform poorly on new data
- Example: If a linear regression model is used to predict housing prices, but the relationship between the input variables and the output variable is non-linear, the model may perform poorly on new data.
- Therefore, choosing the wrong inductive bias (linear instead of non-linear) leads to bad predictions.

# Types Of Biases

- Different learning algorithms vary in the types of biases they employ
- Two useful distinctions among biases are restriction bias and preference bias
- A restriction bias restricts the set of hypotheses considered by the learner  
For example, a learner that considers only conjunctive hypotheses employs a restriction bias
- A preference bias does not restrict the set of hypotheses, but imposes a preference ordering over them.  
For example, a learner that prefers shorter decision trees to longer ones employs a preference bias.”

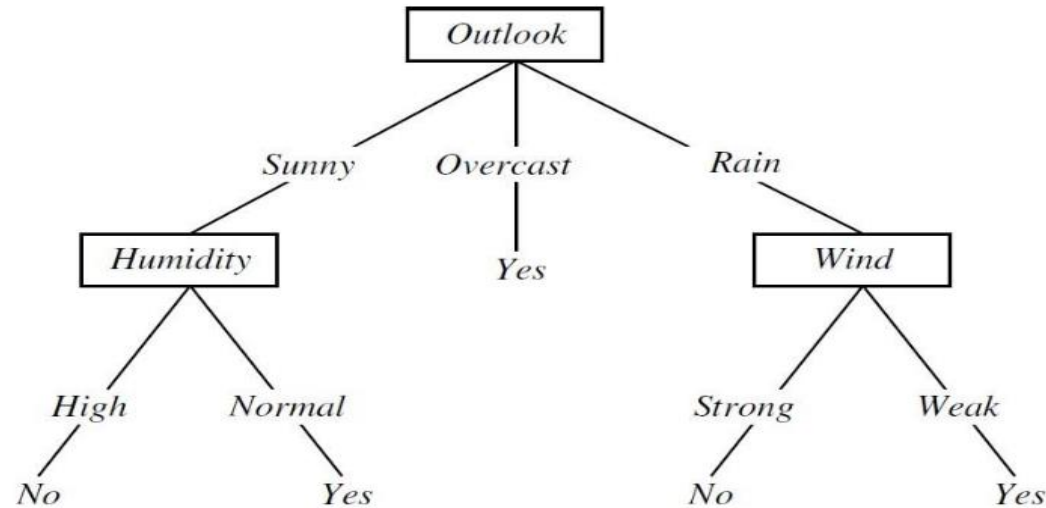
# Inductive Bias Of ID3 Decision Tree (preference Bias)

- For the given training dataset ,we may get multiple number of decision trees
- If only one decision tree is obtained ,there will be no bias in that case
- When there are multiple decision trees which one of these does ID3 choose ?
- **Approximate Inductive Bias(Variable length) :**
  - Shorter trees are preferred over larger trees
- **A closer approximation to the Inductive Bias of ID3(Same length) :**
  - Shorter trees are preferred over larger trees.
  - Trees that place high information gain attributes close to the root are preferred over those that do not

- Why Prefer Short Hypotheses?
- Occam's razor: is the problem-solving principle that the simplest solution tends to be the right one
- “Prefer the simplest hypothesis that fits the data”

| <b>Day</b> | <b>Outlook</b> | <b>Temperature</b> | <b>Humidity</b> | <b>Wind</b> | <b>PlayTennis</b> |
|------------|----------------|--------------------|-----------------|-------------|-------------------|
| <b>D1</b>  | Sunny          | Hot                | High            | Weak        | No                |
| <b>D2</b>  | Sunny          | Hot                | High            | Strong      | No                |
| <b>D3</b>  | Overcast       | Hot                | High            | Weak        | Yes               |
| <b>D4</b>  | Rain           | Mild               | High            | Weak        | Yes               |
| <b>D5</b>  | Rain           | Cool               | Normal          | Weak        | Yes               |
| <b>D6</b>  | Rain           | Cool               | Normal          | Strong      | No                |
| <b>D7</b>  | Overcast       | Cool               | Normal          | Strong      | Yes               |
| <b>D8</b>  | Sunny          | Mild               | High            | Weak        | No                |
| <b>D9</b>  | Sunny          | Cool               | Normal          | Weak        | Yes               |
| <b>D10</b> | Rain           | Mild               | Normal          | Weak        | Yes               |
| <b>D11</b> | Sunny          | Mild               | Normal          | Strong      | Yes               |
| <b>D12</b> | Overcast       | Mild               | High            | Strong      | Yes               |
| <b>D13</b> | Overcast       | Hot                | Normal          | Weak        | Yes               |
| <b>D14</b> | Rain           | Mild               | High            | Strong      | No                |

For this example,  
 There are 4 Attributes, 1 target Attribute(Play Tennis)  
 In total  $2^4$  decision trees are possible  
 Out of 16 decision trees only 1 is used



Root Entropy:  
Yes = 9, No = 5

$$H(S) = - (9/14) \cdot \log_2(9/14) - (5/14) \cdot \log_2(5/14) = 0.94$$

Information Gain at Root:  
Outlook = 0.247 (highest)  
Humidity = 0.152  
Wind = 0.048  
Temperature = 0.029  
□ So root = Outlook

Branch Splits:  
Overcast → all Yes (pure leaf)  
Sunny (2Y/3N) → best split by Humidity  
    High → No, Normal → Yes (pure)  
Rain (3Y/2N) → best split by Wind  
    Strong → No, Weak → Yes (pure)

Why this tree?

- Outlook gives max IG at root.
- Sunny → Humidity and Rain → Wind give perfect classification.
- Overcast is already pure.

Result: short, pure, and optimal tree as per ID3 bias.

# Inductive Bias Of Candidate Elimination (Restriction Bias)

- The Candidate Elimination Algorithm will converge toward the true target concept provided
  - It is given accurate training examples without any errors
  - Its initial hypothesis space contains the target concept
- If target concept is not present in hypothesis space such hypothesis space is called as **Biased Hypothesis Space**
- Example:

| Example | Sky   | AirTemp | Humidity | Wind   | Water | Forecast | EnjoySport |
|---------|-------|---------|----------|--------|-------|----------|------------|
| 1       | Sunny | Warm    | Normal   | Strong | Warm  | Same     | Yes        |
| 2       | Sunny | Warm    | High     | Strong | Warm  | Same     | Yes        |
| 3       | Rainy | Cold    | High     | Strong | Warm  | Change   | No         |
| 4       | Sunny | Warm    | High     | Strong | Cool  | Change   | Yes        |

- In Biased Hypothesis space:
- Suppose the Hypothesis Space  $H$  is restricted to only two attributes:  $\langle \text{Sky}, \text{AirTemp} \rangle$
- Example hypothesis:  $\langle \text{Sunny}, \text{Warm} \rangle$
- But if the true target concept is “EnjoySport when Humidity=Normal”, then this  $H$  cannot express it.
- Candidate Elimination fails (Version Space may become empty)
- This is called **restrictive bias** because the algorithm only searches within the restricted hypothesis space  $H$  and assumes the target concept must be representable within it.

- To overcome this, it is converted to Unbiased Hypothesis space
- Each hypothesis is written as  $\langle \text{Sky, AirTemp, Humidity, Wind, Water, Forecast} \rangle$
- Values can be specific (e.g., Sunny), ? (any), or  $\emptyset$  (no value).
- Example hypotheses:
  - $\langle \text{Sunny, ?, Normal, ?, ?, ?} \rangle \rightarrow \text{EnjoySport if Sky=Sunny AND Humidity=Normal}$
  - $\langle \text{?, Warm, ?, ?, ?, ?} \rangle \rightarrow \text{EnjoySport if AirTemp=Warm}$
- Now, H is large enough to represent many possible concepts  $\rightarrow$  Unbiased

## Positive Impacts

- Inductive bias helps the algorithm focus on the important parts of the data and ignore irrelevant or noisy information
- It makes learning faster and more efficient by guiding the algorithm toward useful patterns
- By avoiding overfitting, inductive bias helps the model generalize better to new, unseen data

## Negative Impacts

- If the chosen bias doesn't match the problem, the algorithm may make inaccurate predictions or fail to capture the correct patterns
- A very strong bias can limit the model's ability to learn more complex relationships that don't fit its assumptions
- Inductive bias can sometimes lead the model to focus too much on certain features, ignoring others that may be important

## References

- Tom M. Mitchell, Machine Learning, McGraw-Hill, 1997
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- GeeksforGeeks, Inductive Bias in Machine Learning



THANK YOU